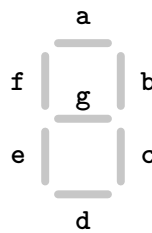
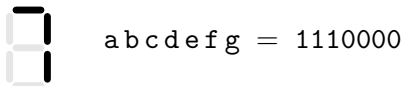


*Answer key***Problem 1.**

A *seven-segment display* shows a digit by lighting up to seven little bars, labelled **a** to **g**. To draw a digit you switch some bars on and leave the rest off — so the whole display is really just **7 bits**: one bit per bar, where 1 means *on* and 0 means *off*. The seven bits for a digit are its row of the display's *truth table*.



Worked example — the digit 7. To make a **7** you light segments **a**, **b** and **c**; every other bar stays off.



$$a b c d e f g = 1110000$$

Read the bars in order **a**, **b**, ..., **g** to get the seven output bits.

Your turn. For each digit below, shade the bars that light up, then write its seven output bits (1 = lit, 0 = off).

digit	display	a	b	c	d	e	f	g
0	0	1	1	1	1	1	1	0
1	1	0	1	1	0	0	0	0
2	2	1	1	0	1	1	0	1
3	3	1	1	1	1	0	0	1
4	4	0	1	1	0	0	1	1
5	5	1	0	1	1	0	1	1
6	6	1	0	1	1	1	1	1
7	7	1	1	1	0	0	0	0
8	8	1	1	1	1	1	1	1
9	9	1	1	1	1	0	1	1

Hint: it helps to picture the digit on the display first, then tick off each of the seven bars in turn.

Problem 2.

Advanced

A display chip can't read the shape "7" — it is handed the digit as a *binary number* and must switch on the right bars. (This is how a digital clock or a scoreboard lights up its digits.) Each digit arrives as binary on the *input*, and the seven segment bits come out as the *output*. So this time **both the input and the output are binary numbers**.

(a) Every digit must reach the display as a binary number. How many bits do you need to represent every digit from 0 to 9? **4 bits** — the largest digit is $9 = 1001$, which already needs 4 bits (3 bits only count up to 7).

(b) Complete the truth table: write each digit as a 4-bit binary *input*, then the seven segment bits (a–g) it switches on as the *output*.

decimal	input (binary)				output (segments)						
	8	4	2	1	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

The seven output columns are exactly the rows you found in Problem 1 — same display, now driven by a binary input.
